Lena Eys

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Education

University of Toronto **Sepember 2024 – May 2029**
Bachelor of Applied Science (B.A.Sc) Mechanical Engineering + PEY Co-op *Toronto, ON*

Experience

Laboratory Technician **May 2026 – August 2026**
Wilfrid Laurier University, Neuromechanics Lab *Kitchener, ON*

- Independently designed and built a prototype of a flip-up tripping device for use in trip perturbation experiments that could be triggered by a combination of both computer and forceplate input.
- Prepared a Bill of Materials; performed torque calculations for the tripping barrier; programmed Arduino controls; designed and integrated electrical circuits on a breadboard; and troubleshot using a multimeter and continuous design iterations.
- Collaborated with manufacturing companies to ensure parts were delivered within required timelines/tolerances.
- All design documents and prototypes were completed in 75% of the allotted timeframe.

University of Toronto Engineering Students Consulting Team Member **October 2025 – April 2026**
University of Toronto, UTESCA *Toronto, ON*

- Worked on UTESCA's Haptic Vision project developing a robotic solution to help the visually impaired interact with digital platforms.
- Utilized Python and Unity to convert 2D user interfaces into a digital 3D model, ready for client integration with existing robotic prototypes.
- The full product went on to win the 2026 Troost ILead Difference Maker Award which celebrates making a positive impact in service to others, earning our client a scholarship of $50,000

Research Assistant **June 2025 – August 2025**
Wilfrid Laurier University, Neuromechanics Lab *Kitchener, ON*

- Analyzed data from in-ground force plates, a load cell and oscilloscope setup, and 3D markers connected to an OptiTrack system.
- Experiments included finding shoe-ground/insole-fabric static and dynamic coefficients of friction, as well as tracking the max moment while a shoe underwent torsion and bending. This allowed the research team to develop a comprehensive set of footwear properties to use in future scientific papers.
- All data was filtered and graphed, and the findings were summarized in a final report.

Maker Lab Intern **July 2024 – August 2024**
Wilfrid Laurier University, Maker Lab *Kitchener, ON*

- Tutored students in Solidworks design and Arduino coding. Within a week of teaching, they had completed CAD training and were able to code movement patterns for a stepper motor.
- Designed the housing of an introductory Arduino coding box aimed at promoting coding for women at a Laurier Women in STEM event.
- Personal projects included using CAD to design a prosthetic wrist, modeling a storage room for renovations, Arduino custom characters, programming a re-centering motor and automatically activated fan.

Extracurricular

University of Toronto Varsity Cross Country Member **September 2024 – April 2026**

- Competed in 6k-8k races with a weekly training commitment of 15-20 hours.

Laurel Creek Track Club competitive member **July 2022 – present**

- Trained and competed in middle distance events (400m-3000m) and cross country (4-8k) in Canadian provincial and national championships, as well as showcases in Ohio and Michigan.
- U18 Steeplechase provincial gold and Legion Nationals bronze medalist (2023).

Technical Skills

Mechanical: Solidworks (CSWA certified), Trimble Sketchup, 3D printing, composites, soldering, machining
Software: Arduino, Python, MATLAB, OptiTrack, Visual 3D, Minitab, Microsoft Excel, Unity, LTspice
Professional: Quantitative research, client consultation, DELF B-2 French Certification